

Lens Design and Characteristics

Laboratory Experiment #1

By

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Submitted To:

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Lab Report

1.) Objective

The objective and intent of our group in this experiment was to design and build an optical system with variable magnification.

2.) Materials

- Optical Table (SG-22-2)
- Damped Rod
- HeNe Laser
- Adjustable Mounting Fixtures for HeNe Laser
 - Laser Mount (ULM-TILT)
 - Rod Clamp (340-RC)
- 4x Post Holders (VPH-2)
- Goggles
- 2x 1" Dia. Adj. Mirror Mount (E55-532)
- 1m Dovetail Optical Rail (E54-402)
- 2x "small" 1.5" Carriers (E55-339)
- 2x "large" 2.5" Carriers (E54-404)
- Lens Kit (E52-305) (DCX +50 & +100 mm; DCV -100 mm)
- 300 um Pin Hole
- 4x B-1 Bases
- 2x Iris Diaphragms
- 2x Post Collar
- Meter Stick and Plastic Calipers
- Screw Kit, $\frac{1}{4}$ x 20 (SK-25A)
- Tape

3.) Background

There are many possible designs for zoom lenses, the most complex ones having upwards of thirty individual lens elements, and multiple moving parts. Most follow the same basic design; generally they consist of a number of individual lenses that may be either fixed, or source axially doing the body of the lenses.

As the magnification of a zoom lens changes, it is necessary to compensate for any movement of the focal plane to keep the focused image sharp. This compensation may be done by mechanical means (moving the complete lens assembly as the magnification of lens changes), or optically (arranging the position focal plane to vary as little as possible as the lens is zoomed).

A simple scheme for a zoom lens divides the assembly into two parts: a focusing lens similar to a standard, fixed-focal-length photographic lens, preceded by an afocal zoom system, an arrangement of fixed and movable lens elements that does not focus the light, but alters the size of a beam of light traveling through it, and thus the overall magnification of the lens system.

4.) Procedure

Lens Design:

1. Unblock laser beam.
2. Tape a 300 μm pin hole to the laser aperture so that the projected beam is brightest (centered).
3. Place a DCX ($f_1 = +100 \text{ mm}$) lens (L1) into a lens holder and slide it onto the first post holder (nearest to the laser) 17.5 cm from the pin hole and record it as d1. As each lens is placed into the lens system, center the beam through the lens so that it is centered on the iris.
4. Mount a post holder to each of the two center carriers using B-1 bases.
5. Mount a DCV ($f_2 = -100 \text{ mm}$) lens (L2) into a lens holder and place it on the second post holder 16.75 cm past the L1 and record it as d2.
6. Mount a DCX ($f_3 = +50 \text{ mm}$) lens (L3) into a lens holder and place it on the third post holder 29.5 cm past L2 lens.
7. Open the iris.
8. Adjust the L3 lens position until the projected image of the pin hole is focused on the wall and record it as d3.
9. Measure the distance d4 between the L3 lens and the wall. It should be 86.25 cm. Record d4.
10. Record y (original size of beam diameter) as 300 μm .
11. Measure the size of the beam on the wall. Label it y' and record it.
12. Move the L2 lens 5 cm (in increments of 0.5 cm) towards the wall.
13. At each step focus the image with the L3 lens (adjust the lens position).
14. At each step measure the beam size on the wall y' and record it. Measure d2, d3, and d4 distances and record them.
15. Calculate the system magnification using the relationship: $M = y' / y$



Lab setup.



Measuring distances.



Making sure beam passes through pin hole



Measuring beam diameter

5.) Data & Observations

TABLE 1

	d1 cm	d2 cm	d3 cm	d4 cm	y um	y' cm	M
1	17.5	16.75	5.5	108.75	300	0.002	M1 = 0.15
2		21.75	4.75	103.5		0.003	M2 = 0.1
3		26.75	4.25	100		0.002	M3 = 0.15
4		31.75	4.25	95.25		0.003	M4 = 0.1
5		36.75	4.25	90		0.004	M5 = 0.075
6		41.75	4.25	85		0.005	M6 = 0.06
7		46.75	4.25	80.25		0.005	M7 = 0.06
8		51.75	4.25	75		0.006	M8 = 0.05
9		56.75	4.25	70		0.006	M9 = 0.05
10		61.75	4.25	66.25		0.007	M10 = 0.043

$$M1 = 300 \text{ um} / 0.002 \text{ cm} \rightarrow 0.15$$

$$M2 = 300 \text{ um} / 0.003 \text{ cm} \rightarrow 0.1$$

$$M3 = 300 \text{ um} / 0.002 \text{ cm} \rightarrow 0.15$$

$$M4 = 300 \text{ um} / 0.003 \text{ cm} \rightarrow 0.1$$

$$M5 = 300 \text{ um} / 0.004 \text{ cm} \rightarrow 0.075$$

$$M6 = 300 \text{ um} / 0.005 \text{ cm} \rightarrow 0.06$$

$$M7 = 300 \text{ um} / 0.005 \text{ cm} \rightarrow 0.06$$

$$M8 = 300 \text{ um} / 0.006 \text{ cm} \rightarrow 0.05$$

$$M9 = 300 \text{ um} / 0.006 \text{ cm} \rightarrow 0.05$$

$$M10 = 300 \text{ um} / 0.007 \text{ cm} \rightarrow 0.043$$

The data obtained is as accurate as possible by measurement tools and the conductors of the experiment. This data shows that the HeNe beam magnification was getting smaller with distance, possibly due to the limited range of movement on the dovetail rail. Some errors can occur during the experiment if the power, measured distance, or calculations are incorrect.

6.) Answers to Lab Questions

1.) What is the range of the system magnification? $0.15 - 0.043 = 0.107$

2.) Is it possible to achieve higher magnification (zoom system) between -85 to -150? In this case, how should the three lenses be arranged? The +50 lens should be in front, negative -100 lens stays in the second position, and the +100 lens should be in the last position.

7.) Conclusion

In conclusion, the laboratory experiment showed that as distance increased, the beam diameter also increased in size due to the beam divergence. The experiment yielded expected results because the magnification decreased with distance, which is like when you try to zoom in on your phone camera and you max out your zoom, the far image appears blurry. The only part of the lab that was unexpected was that the distance we calculated with was 5 centimeters and not 5 centimeters at 0.5 centimeter increments which was the major error to note for this the experiment. Improvements would be to ensure that user errors and equipment errors are at a minimum so that the results are as accurate as possible for future work.